Michael Cho

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Research Interests

Research interests lie at the intersection of biomedical engineering and machine learning, focusing on wearable cardiomechanical sensing, generative modeling of physiological signals, federated and deep learning, edge AI, and data engineering with an emphasis on fairness, ethics, privacy, and security. Expertise includes designing end-to-end machine learning systems, high performance computing, and embedded system architectures utilizing deep learning accelerators (e.g., Coral TPU) for real-time, on-device anomaly detection and classification for wearable health monitoring.

Education

2022 – Present | Ph.D. in Electrical and Computer Engineering | **Georgia Institute of Technology**
Ph.D. Candidate · GPA: 4.0/4.0 | Atlanta, GA

2022 – 2025 | MSc. in Electrical and Computer Engineering | **Georgia Institute of Technology**
GPA: 4.0/4.0 | Atlanta, GA

2020 – 2022 | Bachelor of Science in Computer Science | **University of Washington**
GPA: 3.9/4.0 | Seattle, WA

Experience

Jul 2026 – Dec 2026 | **Applied Machine Learning Scientist** | **WHOOP**
Core Machine Learning Research | Boston, MA

During this co-op, developed machine learning algorithms on wearable physiological data and productionized them for deployment. Built end-to-end ML pipelines for production, spanning data processing, model training and evaluation, and integration into production services.

Jan 2023 – Present | **Graduate Research Assistant** | **Georgia Institute of Technology**
Inan Research Lab · *Advisor: Dr. Omer T. Inan* | Atlanta, GA

Developed and applied Deep Learning algorithms for cardiovascular signal processing (e.g., ECG, SCG, PPG), collaborating with interdisciplinary teams to advance cardiac health research. Curated a large-scale dataset of over 17,000 annotated heartbeats. Leveraging this dataset, developed deep learning signal quality classification models that achieved 25% greater accuracy for signal quality determination of cardiac signals over traditional algorithms. Spearheaded the implementation of on-device deep learning frameworks on edge accelerators (Coral TPU), enabling real-time anomaly detection and few-shot personalization for wearable cardiac signals. Automated annotation workflows to improve ML benchmarking and reproducibility, while integrating HPC clusters and hardware systems to characterize physiological stress responsivity. Contributed to edge-efficient seismocardiogram denoising and geometry-aware generative models for translating cardiovascular signals across wearable sensor locations.

Feb 2023 – Sep 2023 | **Educator Workforce Data Specialist** | **Professional Educator Standards Board**
| Olympia, WA

Orchestrated an organization-wide data restructuring and automated backup initiative, modernizing legacy architecture to ensure 100% data redundancy and improve query efficiency by 37%. Led the rigorous analysis of complex educator workforce datasets, synthesizing quantitative trends to inform state-level policy and drive evidence-based improvements in regulatory standards.

Jun 2021 – Jun 2022 | **Research Assistant** | **University of Washington**
Engineering and Mathematics · *Advisor: Dr. Sunwoong Kim* | Seattle, WA

Developed HELPSE (Homomorphic Encryption-based Lightweight Password Strength Estimation) to enable privacy-preserving analysis on cyber-physical systems. Built a virtual keyboard-based client-server system utilizing a polynomial degree of 32K to capture and encrypt passwords. Implemented numerical algorithms in C++ within the homomorphic encryption domain, achieving a server-side execution time of approximately 5 seconds and an average error rate of less than 1% compared to standard baselines.

Dec 2020 – Jun 2022 **Research Assistant** — **University of Washington**
Ubicomp Lab · *Advisor: Dr. Afra Mashhadi* — Seattle, WA

Implemented Federated Learning (FL) frameworks (TensorFlow Federated, PyTorch) for crowdsourced object detection, achieving 85.13% accuracy across distributed edge devices while ensuring 100% data privacy by keeping raw user data on-device. Optimized model aggregation to outperform local training baselines by 15%, validating the communication efficiency and efficacy of decentralized learning for sensitive data.

Publications

Cho, Michael and Afra Mashhadi (2022). "Caring Without Sharing: A Federated Learning Crowdsensing Framework for Diversifying Representation of Cities". In: *Mobile and Ubiquitous Systems: Computing, Networking and Services*, pp. 601–616. DOI: 10.1007/978-3-030-94822-1_39.

Cho, Michael et al. (2022). "HELPSE: Homomorphic Encryption-based Lightweight Password Strength Estimation in a Virtual Keyboard System". In: *Proceedings of the Great Lakes Symposium on VLSI 2022*. GLSVLSI '22, pp. 405–410. DOI: 10.1145/3526241.3530338.

Abbaraju, Vikram, Ty Lees, et al. (2025). "Characterizing Central-Autonomic Dynamics during an Episodic Memory Task Using Multi-Modal Neural and Cardiomechanical Signals". In: *IEEE Transactions on Biomedical Engineering*, pp. 1–12. DOI: 10.1109/TBME.2025.3622916.

Cho, Michael J., Vikram Abbaraju, et al. (2025). "Quantifying Opioid Withdrawal through Cardio-Mechanical Variability Using Multi-Modal Wearable Sensors". In: *2025 IEEE 21st International Conference on Body Sensor Networks (BSN)*, pp. 1–4. DOI: 10.1109/BSN66969.2025.11337294.

Abbaraju, Vikram, Farhan N. Rahman, et al. (2026). "Towards Monitoring Acute Opioid Withdrawal Severity Using a Multimodal Wearable Chest Patch". In: *ACM Transactions on Computing for Healthcare*. DOI: 10.1145/3833425.

Cho, Michael J., Cem O. Yaldiz, et al. (2026). "Seismocardiography Pig Hypovolemia Dataset for Signal Quality Indexing and Validated Cardiac Timings". In: *Scientific Data*. DOI: 10.1038/s41597-026-06733-2.

Cook, Jacob M. et al. (2026). "SCRIBE Design and Validation: Simultaneous Cardiac and Respiratory Impedance Biomarkers with Electromechanics". In: *IEEE Sensors Journal*. DOI: 10.1109/JSEN.2026.3702932.

Emirdagi, Ahmet R. et al. (2026). "EffSCG: An Efficient Framework for Real-Time Seismocardiogram Denoising on Resource-Constrained Edge Devices". In: *IEEE Journal of Biomedical and Health Informatics*. DOI: 10.1109/JBHI.2026.3718375.

Kilic, Onur Selim et al. (2026). "Cylindrical Geodesic Flow Matching for Quasiperiodic Physiological Signal Transformation". In: *Advances in Neural Information Processing Systems (NeurIPS 2026), Main Conference, Sydney*. arXiv: 2610.08510.

Awards, Honors and Memberships

NSF Graduate Research Fellowship (GRFP)
President's Fellowship, Georgia Institute of Technology
Steve W. Chaddick Fellowship, Georgia Institute of Technology
Mary Gates Research Scholarship, University of Washington
Ronald E. McNair Post-baccalaureate Achievement Program, University of Washington
Officer, Association for Computing Machinery (ACM)
Member, Tau Sigma National Honor Society & MESA (Mathematics Engineering Science Achievement)
USPA, Current Georgia Powerlifting Record Holder for Bench (486lbs), Deadlift (705lbs), and Total (1731lbs)

Teaching and Service

2022 **Graduate Teaching Assistant**, Georgia Tech (ECE 3005): taught technical communication
2022 **Children's Volunteer**, Hands On Atlanta: education and community activities for children
2020 – 2022 **Mentor**, Seattle Central College Reentry Support: mentored formerly incarcerated students
2016 – 2019 **Academic Tutor**, Centralia College: academic support across multiple subjects

Technical Skills & Languages

Languages: Python, C++/C, MATLAB, Bash, Lua, Java, SQL, LaTeX
Data Science: PyTorch, TensorFlow, Pandas, Dask, Plotly, Scikit-learn, NumPy
DevOps & Tools: Docker, Kubernetes, Spark, Slurm (HPC), Proxmox, VMs, Unix, AWS, S3
Spoken Languages: English (Advanced), Spanish (Conversational), Japanese (Basic), Korean (Basic)

Last updated: October 20, 2026